

A Normal Form Preserving Minimality: An Order-4 Counterexample to the Komori–Kashima Problem

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Abstract

A formula is minimal in a logic L if it belongs to L and no strictly more general formula, with respect to the substitution preorder, belongs to L . We show that a polarized extension-variable translation S preserves minimality: if α is minimal in a logic L that contains the axiom $a \rightarrow a$ and the formula $S(\alpha)$, then $S(\alpha)$ is also minimal in L . If L is an implicative logic containing **BCI**, then $S(\alpha) \in L$ whenever $\alpha \in L$, and $L + \alpha = L + S(\alpha)$. Since $S(\alpha)$ has order at most 4, we obtain a minimality-preserving normal form: every implicative logic over **BCI** that is axiomatizable by formulas minimal in classical logic is axiomatizable by such formulas of order at most 4, and the bound 4 is optimal. As a corollary, applying S to the counterexample of Nakamura and Matsuda to the Komori–Kashima problem yields an explicit axiom of Gödel–Dummett logic of order 4 that is minimal in classical logic, which answers their Open Problem 2. We also show that the results extend to extensions of **FL_e** in the multiplicative language and of **FL_{ew}** in the full language. The main theorem and the soundness of S over **BCI** have been formalized in Lean 4.

Keywords: Intermediate logic, Substructural logic, Minimal formulas, Substitution preorder, Axiomatization, Extension variables.

1 Introduction

A formula α is *minimal* in a logic L [12] if $\alpha \in L$ and no formula *strictly more general* than α with respect to the *substitution preorder* belongs to L . For example, Peirce’s law $((a \rightarrow b) \rightarrow a) \rightarrow a$ is *minimal* in classical logic **CL**, whereas $(a \rightarrow b) \rightarrow (a \rightarrow b)$ is not, since it is a proper substitution instance of the classical tautology $p \rightarrow p$.

Since *minimality* depends on the syntactic shape of a formula, transformations of axioms into equivalent ones need not preserve it, and it is not clear a priori whether an axiomatization by *minimal* formulas can be brought into a normal form, for instance one of bounded order, without losing *minimality*.

This question arises in the problem of Komori and Kashima [12, 11], who asked which logics can be axiomatized by formulas *minimal* in **CL**, and in particular whether every intermediate logic axiomatizable in this way is either intuitionistic logic **IL** or **CL**. Nakamura and Matsuda [13] answered the latter question in the negative: they exhibited a formula G' of order 5 that is *minimal* in **CL** and axiomatizes Gödel–Dummett logic **LC** over **IL**, and they asked whether a counterexample of order 4 exists [13, Open Problem 2].

[Plot: Check the formulation of the problem against [12], [11] and [13, Sect. 1].]

1.1 Main results

We show that a translation S based on **extension variables** (Definition 3.1) preserves **minimality**. It replaces every compound subformula occurrence u of α by a fresh variable x_u and records the meaning of x_u in a premise E_u , a one-way definition whose direction depends on the polarity of u :

$$S(\alpha) = E_{u_1} \rightarrow \cdots \rightarrow E_{u_m} \rightarrow x_\varepsilon.$$

The translation is a variant of the polarized translation φ^+ of Jeřábek [3, Def. 3.6], obtained essentially by removing the conjunctions “ $\wedge 1$ ” with which Jeřábek wraps each definition. Since each premise E_u contains exactly two implications, $S(\alpha)$ has order at most 4, and it can be computed in linear time.

Theorem 1.1 (Main Theorem). *Let L be a logic containing the identity axiom $I = a \rightarrow a$. If α is minimal in L and $S(\alpha) \in L$, then $S(\alpha)$ is minimal in L .*

The hypothesis $S(\alpha) \in L$ is satisfied for every $\alpha \in L$, and moreover $L + \alpha = L + S(\alpha)$, whenever L contains the linear implicative logic **BCI** (Corollary 5.2). Consequently, S provides a minimality-preserving normal form for axioms: over **BCI**, every axiomatization by **CL**-minimal formulas can be replaced by one of order at most 4 (Corollary 5.3). The bound 4 is optimal, because classical tautologies of order at most 3 are already intuitionistic theorems (Proposition 5.4). Theorem 1.1, in the generality stated above, and Corollary 5.2 have been formalized in Lean 4 (Appendix B).

The hypotheses on the base logic cannot be weakened substantially. The axiom I cannot be dropped: there is a logic containing **BC** but not I in which the conclusion of Theorem 1.1 fails (Proposition 7.1). Nor can **BCI** be replaced by **CK** or **BK** in Corollary 5.2 (Appendix C).

1.2 Application to the Komori–Kashima problem

Applied to G' , the translation yields an explicit axiom $S(G')$ of **LC** of order 4 that is **minimal** in **CL** (Corollary 6.1). This answers [13, Open Problem 2] affirmatively. Since Corollary 5.3 holds over **BCK**, it also applies to Komori’s original formulation of the problem over **BCK** [13, footnote 1]. Moreover, in the study of which logics are axiomatizable by **CL**-minimal formulas [13, Open Problem 1], one may restrict attention to formulas of order 4 (Corollary 5.3 and Proposition 5.4).

1.3 Substructural logics

The translation and Theorem 1.1 extend to languages with fusion, lattice connectives and constants. The analogue of Corollary 5.2 holds over **FL_e** in the multiplicative language and over **FL_{ew}**, in particular over intuitionistic logic, in the full language. Over **FL_e** with additive connectives, however, neither S nor Jeřábek’s φ^+ serves our purpose (Section 8).

1.4 Overview of the proof

The proof of Theorem 1.1 is purely syntactic; it uses only closure under substitution and modus ponens, together with the axiom I . It proceeds in two steps. First, a formula of L is **minimal** in L if and only if no **one-step generalization** of it belongs to L , where a **one-step generalization** is obtained by replacing some occurrences of a single subformula by a fresh

variable (Proposition 2.2). Second, every one-step generalization α' of $S(\alpha)$ can be decoded into a one-step generalization β of α : there is a substitution θ such that

$$\theta\alpha' = (\gamma_1 \rightarrow \gamma_1) \rightarrow \cdots \rightarrow (\gamma_k \rightarrow \gamma_k) \rightarrow \beta$$

(Lemma 4.2). Hence, if α' belonged to L , then so would β , contradicting the minimality of α . Polarity plays no role in this argument; it is needed only to ensure that $S(\alpha) \in L$. Example 4.4 illustrates the decoding for Peirce’s law.

1.5 Related work

The translation S is built from well-known ingredients. Extension variables go back to Tseitin [8], and Statman [14] used them to reduce intuitionistic propositional logic to its implicational fragment (see also Švejdar [7]). One-way definitions chosen according to polarity appear in the Plaisted–Greenbaum translation [5] and in Mints’ normal form [4], and Rybakov [6] uses extension variables in the study of admissible rules.

Our translation is closest to that of Jeřábek [3]. His translation φ^+ introduces a fresh variable for every occurrence of a non-variable subformula, uses the same polarized one-way definitions, and wraps each of them as $E_u \wedge 1$. The translation S differs from φ^+ only in that it omits these wrappers, carries the product of the definitions into an iterated implication, and treats constants as atoms. With the wrappers, $\mathbf{FL}_e + \varphi = \mathbf{FL}_e + \varphi^+$ holds for every formula φ in the full language [3, Thm. 3.8]; Jeřábek uses this to show that the substructural hierarchy of Ciabattoni, Galatos and Terui [1] collapses to the level $\mathcal{N}_3$ over $\mathbf{FL}_e$. The main point of the present paper is the observation that, without the wrappers, the translation preserves minimality; to our knowledge, this has not been observed before. The wrappers destroy minimality (Section 8).

[Plot: Check the exact forms used in [14], [7], [4].]

1.6 Organization of the paper

Section 2 fixes notation and terminology, Section 3 defines the translation S , and Section 4 proves Theorem 1.1. Section 5 establishes the basic properties of S over $\mathbf{BCI}$ and derives the normal form, and Section 6 applies it to the Komori–Kashima problem. Section 7 shows that the axiom I cannot be dropped from Theorem 1.1, and Section 8 extends the results to substructural logics. Section 9 concludes with open problems. Appendix A proves Proposition 2.2, Appendix B describes the Lean formalization, and Appendix C shows that B and C cannot be dropped from Corollary 5.2.

2 Preliminaries

2.1 Formulas and logics

Except in Section 8, we work in the implicational language: formulas are built from variables by the implication $\rightarrow$. As usual, implication associates to the right: $\alpha \rightarrow \beta \rightarrow \gamma$ stands for $\alpha \rightarrow (\beta \rightarrow \gamma)$. The *order* $\text{Ord}(\alpha)$ of a formula α is defined by $\text{Ord}(p) = 1$ for variables p and $\text{Ord}(\alpha \rightarrow \beta) = \max\{\text{Ord}(\alpha) + 1, \text{Ord}(\beta)\}$, and its *height* by $h(p) = 0$ and $h(\alpha \rightarrow \beta) = 1 + \max\{h(\alpha), h(\beta)\}$.

A *substitution* σ is a map from variables to formulas, and $\sigma\alpha$ denotes the result of applying σ to α . A *logic* is a set of formulas that is closed under modus ponens and substitution. For a set F of formulas, F^* denotes the smallest logic containing F , and for a logic L we write $L + F$ for $(L \cup F)^*$ and $L + \alpha$ for $L + \{\alpha\}$. We shall repeatedly use the following closure properties of a logic L :

- (C1) if $\gamma \in L$, then $\sigma\gamma \in L$ for every substitution σ ;
- (C2) if $\gamma_1 \rightarrow \dots \rightarrow \gamma_k \rightarrow \beta \in L$ and $\gamma_1, \dots, \gamma_k \in L$, then $\beta \in L$;
- (C3) $\gamma \rightarrow \gamma \in L$ for every formula γ .

Every logic satisfies (C1) and (C2), and by (C1) a logic satisfies (C3) if and only if it contains the identity axiom $I = a \rightarrow a$.

2.2 The substitution preorder

⌈ We write $\beta \leq_{\text{sub}} \alpha$ if $\alpha = \sigma\beta$ for some substitution σ , and $\beta <_{\text{sub}} \alpha$ if $\beta \leq_{\text{sub}} \alpha$ but not $\alpha \leq_{\text{sub}} \beta$. The relation $\leq_{\text{sub}}$ is the *substitution preorder*. If $\beta <_{\text{sub}} \alpha$, we say that β is *strictly more general* than α .

⌈ **Definition 2.1.** A formula α is *minimal in* a logic L if $\alpha \in L$ and there is no $\beta \in L$ with $\beta <_{\text{sub}} \alpha$.

2.3 Occurrences and one-step generalizations

⌈ We identify an occurrence of a subformula in a formula α with its position in the syntax tree of α , that is, with a finite word over $\{0, 1\}$. The empty word ε denotes the root, and if the subformula at u is $\gamma \rightarrow \delta$, then γ is at $u0$ and δ is at $u1$. We write α_u for the subformula of α at u . An occurrence u is *strictly below* v if v is a proper prefix of u . An occurrence u is a *leaf* if α_u is a variable, and a *compound occurrence* otherwise. Note that two distinct occurrences of the same formula are never strictly below one another.

⌈ Let Q be a set of occurrences in α such that no element of Q is strictly below another, and let z be a variable. We write $\alpha[Q:=z]$ for the formula obtained from α by replacing the subformula at each $u \in Q$ by z .

⌈ A *one-step generalization* of a formula α is a formula $\alpha[Q:=z]$, where Q is a nonempty set of occurrences of a single formula δ in α , the variable z does not occur in α , and, if δ is a variable, some occurrence of δ does not belong to Q (replacing all occurrences of a variable merely renames it).

Proposition 2.2. *Let L be a logic and let $\alpha \in L$. Then α is minimal in L if and only if no one-step generalization of α belongs to L .*

This is essentially [13, Props. 3.1 and 3.2]; for completeness, we give a proof in Appendix A. [\[Plot: Check whether \[13, Props. 3.1 and 3.2\] cover non-simple formulas.\]](#)

3 The Translation

- ⌈ Fix a formula α . We assign a *polarity*, *positive* or *negative*, to each occurrence in α as follows. The root ε is *positive*. If $\alpha_u = \alpha_{u0} \rightarrow \alpha_{u1}$, then $u1$ has the same polarity as u and $u0$ the opposite one. For each compound occurrence u we choose a new variable x_u , called an *extension variable*, so that distinct occurrences receive distinct variables and no x_u occurs in α . For every occurrence u let

$$r_u = \begin{cases} \alpha_u, & \text{if } u \text{ is a leaf,} \\ x_u, & \text{if } u \text{ is a compound occurrence.} \end{cases}$$

- ⌈ For every compound occurrence u we call $b_u = r_{u0} \rightarrow r_{u1}$ the *body* of u , and we define the *defining premise* of u by

$$E_u = \begin{cases} b_u \rightarrow x_u, & \text{if } u \text{ is positive,} \\ x_u \rightarrow b_u, & \text{if } u \text{ is negative.} \end{cases}$$

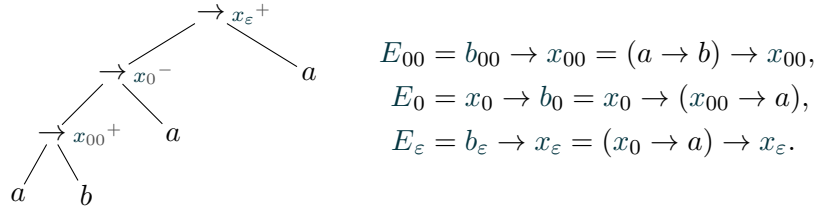
- ⌈ **Definition 3.1.** Let $u_1, \dots, u_m$ be the compound occurrences of α , listed in post-order; that is, u_i precedes u_j if and only if u_i is strictly below u_j , or $u_i = w0u'$ and $u_j = w1u''$ for some words w, u', u'' . The *translation* of α is the formula

$$S(\alpha) = E_{u_1} \rightarrow \dots \rightarrow E_{u_m} \rightarrow r_\varepsilon.$$

- ⌈ If α is a variable, then $m = 0$ and $S(\alpha) = \alpha$. Let θ_α be the substitution that maps each x_u to α_u and fixes all other variables; then $\theta_\alpha(r_v) = \alpha_v$ for every occurrence v of α .

In the following examples, each implication node of the syntax tree of α , that is, each compound occurrence u , is annotated with x_u and the polarity of u ; the left and right children of u are $u0$ and $u1$.

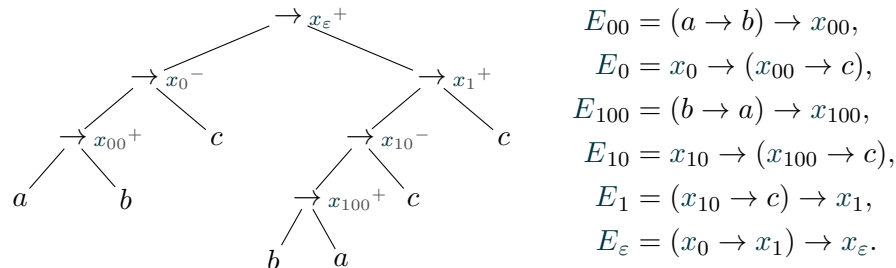
Example 3.2. For Peirce's law $\alpha = ((a \rightarrow b) \rightarrow a) \rightarrow a$:



Hence

$$S(\alpha) = ((a \rightarrow b) \rightarrow x_{00}) \rightarrow (x_0 \rightarrow x_{00} \rightarrow a) \rightarrow ((x_0 \rightarrow a) \rightarrow x_\varepsilon) \rightarrow x_\varepsilon.$$

Example 3.3. For the axiom $\alpha = ((a \rightarrow b) \rightarrow c) \rightarrow ((b \rightarrow a) \rightarrow c) \rightarrow c$ of Gödel–Dummett logic:



Listing the defining premises in this order (post-order), we obtain

$$\begin{aligned} S(\alpha) = & ((a \rightarrow b) \rightarrow x_{00}) \rightarrow (x_0 \rightarrow x_{00} \rightarrow c) \rightarrow ((b \rightarrow a) \rightarrow x_{100}) \rightarrow (x_{10} \rightarrow x_{100} \rightarrow c) \\ & \rightarrow ((x_{10} \rightarrow c) \rightarrow x_1) \rightarrow ((x_0 \rightarrow x_1) \rightarrow x_\varepsilon) \rightarrow x_\varepsilon. \end{aligned}$$

4 Proof of the Main Theorem

Throughout this section, α is a formula that is not a variable, $u_1, \dots, u_m$ are its **compound occurrences** in post-order, and z is a variable not occurring in $S(\alpha)$. By Proposition 2.2, it suffices to show, for a logic L containing I , that if a one-step generalization of $S(\alpha)$ belongs to L , then so does one of α .

□ We first name some occurrences in $S(\alpha)$. For $1 \leq i \leq m+1$, let $T_i = E_{u_i} \rightarrow \dots \rightarrow E_{u_m} \rightarrow x_\varepsilon$; thus $T_1 = S(\alpha)$, $T_{m+1} = x_\varepsilon$, and T_i occurs in $S(\alpha)$ at 1^{i-1} . For $u = u_i$, the formula E_u occurs at **prem**(u) = $1^{i-1}0$, and we let **body**(u) and **def**(u) be the occurrences of b_u and x_u in E_u ; that is, **body**(u) = **prem**(u)0 and **def**(u) = **prem**(u)1 if u is positive, and **body**(u) = **prem**(u)1 and **def**(u) = **prem**(u)0 if u is negative. Finally, for an occurrence v of α we define the occurrence **rep**(v) of r_v in $S(\alpha)$ by **rep**($w0$) = **body**(w)0, **rep**($w1$) = **body**(w)1 and **rep**(ε) = 1^m .

Lemma 4.1.

- (a) Every occurrence in $S(\alpha)$ is exactly one of 1^{i-1} ($1 \leq i \leq m$), **prem**(u), **body**(u), **def**(u) and **rep**(v), and the subformula of $S(\alpha)$ there is T_i , E_u , b_u , x_u and r_v , respectively.
- (b) For a compound occurrence u of α , the variable x_u occurs in $S(\alpha)$ exactly at **def**(u) and **rep**(u). A variable c of α occurs in $S(\alpha)$ exactly at the occurrences **rep**(v) with v a leaf of α and $\alpha_v = c$.
- (c) $h(b_u) = 1$, $h(E_u) = 2$ and $h(T_i) \geq 3$ for $i \leq m$. Hence a formula of height 1 occurs in $S(\alpha)$ only at occurrences **body**(u), a formula of height 2 only at occurrences **prem**(u), and a formula of height at least 3 only at occurrences 1^{i-1} .
- (d) If $b_u = b_v$, then $\alpha_u = \alpha_v$. If $E_u = E_v$, then $u = v$. If $T_i = T_j$, then $i = j$.

Proof. (a) and (b) follow from Definition 3.1, since the **extension variables** are distinct and do not occur in α .

(c) We have $b_u = r_{u0} \rightarrow r_{u1}$ with r_{u0} and r_{u1} variables, E_u is $b_u \rightarrow x_u$ or $x_u \rightarrow b_u$, and $T_i = E_{u_i} \rightarrow T_{i+1}$. The second claim follows from (a), since the formulas at **def**(u) and **rep**(v) are variables.

(d) If $b_u = b_v$, then $\alpha_u = \theta_\alpha(b_u) = \theta_\alpha(b_v) = \alpha_v$. Since b_u is not a variable, E_u is of the form $\gamma \rightarrow p$ with p a variable if and only if u is positive. Hence $E_u = E_v$ implies that u and v have the same polarity, so $x_u = x_v$ and $u = v$. Finally, if $i < j$, then T_j is a proper subformula of T_i , so $T_i \neq T_j$. □

□ We call a one-step generalization $S(\alpha)[P:=z]$ **basic** if every element of P is of the form **body**(u) or **rep**(v).

Lemma 4.2 (Decoding). *Let $\alpha' = S(\alpha)[P:=z]$ be a basic one-step generalization of $S(\alpha)$, and let*

$$Q = \{v \mid \text{rep}(v) \in P\} \cup \{u \mid \text{body}(u) \in P\}.$$

Then $\beta = \alpha[Q:=z]$ is a one-step generalization of α , and there are a substitution θ and formulas $\gamma_1, \dots, \gamma_m$ such that

$$\theta\alpha' = (\gamma_1 \rightarrow \gamma_1) \rightarrow \dots \rightarrow (\gamma_m \rightarrow \gamma_m) \rightarrow \beta.$$

Proof. Let δ be the formula such that P is a set of occurrences of δ . We first show that β is a one-step generalization of α . Clearly $Q \neq \emptyset$, and z does not occur in α , since every variable of α occurs in $S(\alpha)$. If δ is a variable of α , then by Lemma 4.1(b), Q is a set of leaves v with $\alpha_v = \delta$, and some such leaf is not in Q , since P does not contain all occurrences of δ . If $\delta = x_w$, then $P = \{\text{rep}(w)\}$ by Lemma 4.1(b), since P is basic; hence $Q = \{w\}$, and α_w is not a variable. If δ is not a variable, then Q is the set of u with $\text{body}(u) \in P$, so $b_u = \delta$ and $\alpha_u = \theta_\alpha(\delta)$ for all $u \in Q$.

Since P is basic, no occurrence 1^{i-1} , $\text{prem}(u)$ or $\text{def}(u)$ belongs to P . In particular, for every compound occurrence v of α ,

$$\alpha'_{\text{body}(v)} = \begin{cases} z, & \text{if } \text{body}(v) \in P, \\ \alpha'_{\text{rep}(v0)} \rightarrow \alpha'_{\text{rep}(v1)}, & \text{otherwise,} \end{cases} \quad (1)$$

where $\alpha'_{\text{rep}(vj)}$ is z or r_{vj} for $j = 0, 1$. Let $\theta(p) = p$ for every variable p that is not an extension variable, and let

$$\theta(x_v) = \theta(\alpha'_{\text{body}(v)})$$

for every compound occurrence v of α . By (1), the only extension variables that may occur in $\alpha'_{\text{body}(v)}$ are x_{v0} and x_{v1} , so this defines θ by induction on $h(\alpha_v)$.

We claim that $\theta(\alpha'_{\text{rep}(v)}) = \beta_v$ for every occurrence v of α that is not strictly below an element of Q . Here $\text{rep}(v)$ is an occurrence in α' , since no proper prefix of $\text{rep}(v)$ belongs to P : the proper prefixes are of the form 1^{i-1} , together with $\text{prem}(w)$ and $\text{body}(w)$ if $v = wj$, and in the latter case $\text{body}(w) \notin P$ because $w \notin Q$. We argue by induction on $h(\alpha_v)$. If $\text{rep}(v) \in P$, then $v \in Q$ and both sides are z . Otherwise $\alpha'_{\text{rep}(v)} = r_v$. If v is a leaf, then $v \notin Q$ and both sides are α_v . If v is a compound occurrence and $\text{body}(v) \in P$, then $v \in Q$ and both sides are z . Otherwise $v \notin Q$, so neither $v0$ nor $v1$ is strictly below an element of Q , and by (1) and the induction hypothesis,

$$\theta(x_v) = \theta(\alpha'_{\text{rep}(v0)}) \rightarrow \theta(\alpha'_{\text{rep}(v1)}) = \beta_{v0} \rightarrow \beta_{v1} = \beta_v.$$

Since no occurrence 1^{i-1} belongs to P , we have $\alpha' = \alpha'_{\text{prem}(u_1)} \rightarrow \dots \rightarrow \alpha'_{\text{prem}(u_m)} \rightarrow \alpha'_{\text{rep}(\varepsilon)}$. Since $\text{def}(u_i) \notin P$, the formula $\alpha'_{\text{prem}(u_i)}$ is $\alpha'_{\text{body}(u_i)} \rightarrow x_{u_i}$ or $x_{u_i} \rightarrow \alpha'_{\text{body}(u_i)}$, and θ maps it to $\gamma_i \rightarrow \gamma_i$ with $\gamma_i = \theta(x_{u_i})$. By the claim for $v = \varepsilon$, θ maps $\alpha'_{\text{rep}(\varepsilon)}$ to β . $\square$

Lemma 4.3 (Reduction). *If some one-step generalization of $S(\alpha)$ belongs to a logic L , then so does some basic one.*

Proof. Let $\alpha' = S(\alpha)[P:=z] \in L$, where P is a set of occurrences of a formula δ . Since $\sigma\alpha' \in L$ for every substitution σ by (C1), it suffices to find σ such that $\sigma\alpha'$ is a basic one-step generalization of $S(\alpha)$. We distinguish cases on δ .

If δ is a variable of α or $h(\delta) = 1$, then every element of P is of the form $\text{rep}(v)$ or $\text{body}(u)$, respectively, by Lemma 4.1(b) and (c), so α' itself is basic.

If $\delta = x_u$, then $P = \{\text{rep}(u)\}$ or $P = \{\text{def}(u)\}$ by Lemma 4.1(b), since P does not contain all occurrences of δ . In the first case α' is basic. In the second case, the substitution that interchanges x_u and z maps α' to $S(\alpha)[\{\text{rep}(u)\}:=z]$.

If $h(\delta) = 2$, then $\delta = E_u$ and $P = \{\text{prem}(u)\}$ for some u by Lemma 4.1(c) and (d). Let $d \in \{0, 1\}$ be such that $\text{def}(u) = \text{prem}(u)d$. Substituting $E_u[\{d\}:=z]$ for z , we obtain $S(\alpha)[\{\text{def}(u)\}:=z]$, which is treated in the previous case.

If $h(\delta) \geq 3$, then $\delta = T_i$ and $P = \{1^{i-1}\}$ for some $i \leq m$ by Lemma 4.1(c) and (d). Substituting $T_i[\{1^{m-i+1}\}:=z]$ for z , we obtain $S(\alpha)[\{\text{rep}(\varepsilon)\}:=z]$, which is basic. $\square$

Proof of Theorem 1.1. If α is a variable, then $S(\alpha) = \alpha$ and there is nothing to prove. Otherwise, suppose that some one-step generalization of $S(\alpha)$ belongs to L . By Lemma 4.3, some basic one, say α' , belongs to L . Take β , θ and $\gamma_1, \dots, \gamma_m$ as in Lemma 4.2. Then $\theta\alpha' \in L$ by (C1) and $\gamma_i \rightarrow \gamma_i \in L$ by (C3), so $\beta \in L$ by (C2). This contradicts the minimality of α by Proposition 2.2, since β is a one-step generalization of α . Hence no one-step generalization of $S(\alpha)$ belongs to L , and since $S(\alpha) \in L$, $S(\alpha)$ is minimal in L by Proposition 2.2. $\square$

Example 4.4. Let α be Peirce's law as in Example 3.2, and replace the occurrence of x_{00} in E_0 , that is, at $\text{rep}(00)$, by z :

$$\alpha' = ((a \rightarrow b) \rightarrow x_{00}) \rightarrow (x_0 \rightarrow z \rightarrow a) \rightarrow ((x_0 \rightarrow a) \rightarrow x_\varepsilon) \rightarrow x_\varepsilon.$$

Then $Q = \{00\}$ and $\beta = (z \rightarrow a) \rightarrow a$. The substitution θ maps x_{00} to $a \rightarrow b$, x_0 to $z \rightarrow a$ and x_ε to $(z \rightarrow a) \rightarrow a = \beta$, so that

$$\theta\alpha' = ((a \rightarrow b) \rightarrow (a \rightarrow b)) \rightarrow ((z \rightarrow a) \rightarrow (z \rightarrow a)) \rightarrow (\beta \rightarrow \beta) \rightarrow \beta.$$

Since β is not a classical tautology, neither is α' .

5 A Minimality-Preserving Normal Form

$\lceil$ We now apply Theorem 1.1 to logics containing **BCI**. We use the axiom schemes

$$\begin{aligned} B &= (b \rightarrow c) \rightarrow (a \rightarrow b) \rightarrow a \rightarrow c, & B' &= (a \rightarrow b) \rightarrow (b \rightarrow c) \rightarrow a \rightarrow c, \\ C &= (a \rightarrow b \rightarrow c) \rightarrow b \rightarrow a \rightarrow c, & S &= (a \rightarrow b \rightarrow c) \rightarrow (a \rightarrow b) \rightarrow a \rightarrow c, \\ W &= (a \rightarrow a \rightarrow b) \rightarrow a \rightarrow b, & K &= a \rightarrow b \rightarrow a, \end{aligned}$$

and the logics $\mathbf{BCI} = \{B, C, I\}^*$, $\mathbf{BCK} = \{B, C, K\}^*$, $\mathbf{IL} = \{K, S\}^*$, $\mathbf{CL} = \mathbf{IL} + ((a \rightarrow b) \rightarrow a) \rightarrow a$ and Gödel-Dummett logic [9] $\mathbf{LC} = \mathbf{IL} + ((a \rightarrow b) \rightarrow c) \rightarrow ((b \rightarrow a) \rightarrow c) \rightarrow c$. A formula belongs to $\mathbf{CL}$ if and only if it is true under every Boolean valuation.

[\[Plot: Check that the implicational axiom of LC agrees with \[13\].\]](#)

Lemma 5.1 (Soundness). *If $L \supseteq \mathbf{BCI}$, then $\alpha \rightarrow S(\alpha) \in L$ for every formula α .*

Proof. We argue in the implicational fragment of the sequent calculus for **FL_e**, which is complete for **BCI** (see [2]). For a compound occurrence u , let Γ_u be the multiset of the defining premises E_v such that $v = u$ or v is strictly below u ; for a leaf u , let Γ_u be empty. We show, by induction from the leaves upward, that

$$\Gamma_u, \alpha_u \vdash r_u \quad \text{if } u \text{ is positive,} \quad \Gamma_u, r_u \vdash \alpha_u \quad \text{if } u \text{ is negative.}$$

For leaves, these sequents are axioms. Suppose that u is positive and compound. Then $u0$ is negative and $u1$ is positive, so the induction hypotheses and the rule $(\rightarrow L)$ yield $\Gamma_{u0}, \Gamma_{u1}, \alpha_u, r_{u0} \vdash r_{u1}$. Applying $(\rightarrow R)$ and then $(\rightarrow L)$ with $E_u = (r_{u0} \rightarrow r_{u1}) \rightarrow x_u$, we obtain $\Gamma_u, \alpha_u \vdash x_u$. Negative occurrences are treated symmetrically. Every formula of Γ_u is used exactly once, so no structural rule other than exchange is needed. Finally, taking $u = \varepsilon$ gives $E_{u_1}, \dots, E_{u_m}, \alpha \vdash r_\varepsilon$, and hence $\vdash \alpha \rightarrow S(\alpha)$ by exchange and $(\rightarrow R)$. $\square$

Corollary 5.2. *Let L be a logic containing **BCI**, such as **BCI**, **BCK**, **IL**, **LC** or **CL**. Then $L + \alpha = L + S(\alpha)$ for every formula α . If, moreover, α is minimal in L , then so is $S(\alpha)$.*

Proof. By Lemma 5.1, $S(\alpha) \in L + \alpha$. Conversely, the substitution θ_α of Section 3 maps every defining premise E_u to $\alpha_u \rightarrow \alpha_u$ and r_ε to α , so $\alpha \in L + S(\alpha)$ by (C1)–(C3). If α is minimal in L , then $\alpha \in L$, hence $S(\alpha) \in L$ by Lemma 5.1, and Theorem 1.1 applies. $\square$

Corollary 5.3 (Order-4 normal form). *Let L_0 be a logic containing **BCI**. If $L = L_0 + \Gamma$, where every $\alpha \in \Gamma$ is minimal in **CL**, then $L = L_0 + \{S(\alpha) \mid \alpha \in \Gamma\}$, and every $S(\alpha)$ is minimal in **CL** and of order at most 4.*

Proof. The first two claims follow from Corollary 5.2 applied to L_0 and to **CL**. Since every body has order 2, every defining premise has order at most 3, and hence $\text{Ord}(S(\alpha)) \leq 4$. $\square$

The bound 4 cannot be improved: by the next proposition, a classical tautology of order at most 3 never axiomatizes a proper extension of **IL**, whereas **LC** is axiomatized over **IL** by a formula minimal in **CL** (Corollary 6.1).

Proposition 5.4. *Every formula of **CL** of order at most 3 belongs to **IL**.*

Proof. A formula of order at most 2 has the form $p_1 \rightarrow \dots \rightarrow p_k \rightarrow q$ with variables $p_1, \dots, p_k, q$, and we read it as the rule “from $p_1, \dots, p_k$ infer q ”. A formula of order at most 3 has the form $F = H_1 \rightarrow \dots \rightarrow H_n \rightarrow r$, where the H_i are of order at most 2 and r is a variable. Let V be the closure of the empty set under the rules $H_1, \dots, H_n$. The valuation that makes exactly the variables in V true satisfies every H_i , so $r \in V$ because $F \in \mathbf{CL}$. Following the construction of V step by step, we can derive r from $H_1, \dots, H_n$ by modus ponens in **IL**. Hence $F \in \mathbf{IL}$. $\square$

6 Application to the Komori–Kashima Problem

Applying the translation to the formula G' of Nakamura and Matsuda [13], we obtain an affirmative answer to their Open Problem 2.

6.1 An axiom of Gödel–Dummett logic of order 4

Corollary 6.1. *Let G' be the formula of [13, Sect. 4]:*

$$G' = (a \rightarrow a') \rightarrow (b' \rightarrow b) \rightarrow (c'' \rightarrow c') \rightarrow (c''' \rightarrow c') \rightarrow ((a \rightarrow b) \rightarrow c'') \rightarrow ((b' \rightarrow a') \rightarrow c''') \\ \rightarrow (((a' \rightarrow b') \rightarrow c'') \rightarrow ((b \rightarrow a) \rightarrow c''')) \rightarrow c' \rightarrow c \rightarrow c.$$

Then $S(G')$ is the formula

$$S(G') = (x_2 \rightarrow a \rightarrow a') \rightarrow (x_4 \rightarrow b' \rightarrow b) \rightarrow (x_6 \rightarrow c'' \rightarrow c') \rightarrow (x_8 \rightarrow c''' \rightarrow c') \\ \rightarrow ((a \rightarrow b) \rightarrow x_{11}) \rightarrow (x_{10} \rightarrow x_{11} \rightarrow c'') \rightarrow ((b' \rightarrow a') \rightarrow x_{14}) \rightarrow (x_{13} \rightarrow x_{14} \rightarrow c''') \\ \rightarrow (x_{18} \rightarrow a' \rightarrow b') \rightarrow ((x_{18} \rightarrow c'') \rightarrow x_{17}) \rightarrow (x_{21} \rightarrow b \rightarrow a) \rightarrow ((x_{21} \rightarrow c''') \rightarrow x_{20}) \\ \rightarrow (x_{22} \rightarrow c' \rightarrow c) \rightarrow (x_{19} \rightarrow x_{20} \rightarrow x_{22}) \rightarrow (x_{16} \rightarrow x_{17} \rightarrow x_{19}) \rightarrow ((x_{16} \rightarrow c) \rightarrow x_{15}) \\ \rightarrow ((x_{13} \rightarrow x_{15}) \rightarrow x_{12}) \rightarrow ((x_{10} \rightarrow x_{12}) \rightarrow x_9) \rightarrow ((x_8 \rightarrow x_9) \rightarrow x_7) \\ \rightarrow ((x_6 \rightarrow x_7) \rightarrow x_5) \rightarrow ((x_4 \rightarrow x_5) \rightarrow x_3) \rightarrow ((x_2 \rightarrow x_3) \rightarrow x_1) \rightarrow x_1,$$

which has 22 premises, 30 variables and 67 occurrences of variables. It satisfies $\mathbf{IL} + S(G') = \mathbf{LC}$, it is minimal in $\mathbf{CL}$, and $\text{Ord}(S(G')) = 4$. Consequently, there exists a proper intermediate logic that is axiomatizable by a formula of order 4 minimal in $\mathbf{CL}$; that is, [13, Open Problem 2] has an affirmative answer, and the answer to the Komori–Kashima problem is negative already for formulas of order 4.

Proof. By [13, Lemma 4.2], G' is minimal in $\mathbf{CL}$ and $\mathbf{IL} + G' = \mathbf{LC}$. The claims now follow from Corollaries 5.2 and 5.3, except that $\text{Ord}(S(G')) \geq 4$, which holds by Proposition 5.4, since $S(G') \in \mathbf{CL} \setminus \mathbf{IL}$. The displayed formula is obtained from Definition 3.1, where the implication occurrences of G' are numbered $1, \dots, 22$ in pre-order and x_i denotes the extension variable of occurrence i . $\square$

[Plot: The Lean file checks that the displayed formula equals `statman Gprime 8` and has order 4. The table of the defining premises is generated by `paper/gen_SGprime.py` (supplementary material).]

6.2 Top-level premises of order 3

The formula $S(G')$ is rather long. We show that a counterexample of order 4 must have at least two top-level premises of order 3, which partly explains this. For a formula $F = A_1 \rightarrow \dots \rightarrow A_n \rightarrow r$ with r a variable, we call $A_1, \dots, A_n$ the *top-level premises* of F .

Proposition 6.2. *Let $F = A_1 \rightarrow \dots \rightarrow A_n \rightarrow r \in \mathbf{CL}$, where r is a variable and $\text{Ord}(F) \leq 4$. If at most one A_i has order 3, then either $F \in \mathbf{IL}$ or $\mathbf{IL} + F = \mathbf{CL}$.*

Note that minimality of F is not assumed.

Proof. As in the proof of Proposition 5.4, we call a formula $p_1 \rightarrow \dots \rightarrow p_k \rightarrow q$ with variables $p_1, \dots, p_k, q$ ($k \geq 0$) a *Horn clause* and read it as the rule “from $p_1, \dots, p_k$ infer q ”; these are exactly the formulas of order at most 2. Since $\text{Ord}(F) \leq 4$, every A_i has order at most 3. Recall that premises can be permuted in $\mathbf{IL}$.

Let Γ be the set of top-level premises of order at most 2. For a set X of variables, let $\text{Cl}(X)$ be the closure of X under the rules in Γ . We identify a set of variables with the valuation that makes exactly those variables true. Then:

(H1) $\text{Cl}(X)$ satisfies Γ ;

(H2) if $q \in \text{Cl}(X)$, then $\Gamma, X \vdash_{\mathbf{IL}} q$, by modus ponens along the construction of $\text{Cl}(X)$. In particular, if $q \in \text{Cl}(\{p_1, \dots, p_k\})$, then $\Gamma \vdash_{\mathbf{IL}} p_1 \rightarrow \dots \rightarrow p_k \rightarrow q$.

Case 1: no top-level premise has order 3. Then F is $\Gamma \rightarrow r$. By (H1) and $F \in \mathbf{CL}$, we have $r \in \text{Cl}(\emptyset)$, and hence $F \in \mathbf{IL}$ by (H2).

Case 2: exactly one top-level premise, say $A = B_1 \rightarrow \dots \rightarrow B_m \rightarrow s$, has order 3. Then each B_j has order at most 2. Assume that $F \notin \mathbf{IL}$, and put $V_0 = \text{Cl}(\emptyset)$. We first note two facts.

- $r \notin V_0$, since otherwise $\Gamma \vdash_{\mathbf{IL}} r$ by (H2).
- Some B_j is not derivable from Γ in $\mathbf{IL}$. Otherwise $\Gamma, A \vdash_{\mathbf{IL}} s$. Moreover, the valuation $\text{Cl}(\{s\})$ satisfies Γ and A , so $r \in \text{Cl}(\{s\})$ because $F \in \mathbf{CL}$. Then $\Gamma, s \vdash_{\mathbf{IL}} r$ by (H2), and therefore $F \in \mathbf{IL}$, a contradiction.

Fix such a $B_j = p_1 \rightarrow \dots \rightarrow p_k \rightarrow q$, and put $V_1 = \text{Cl}(\{p_1, \dots, p_k\})$. Then $V_0 \subseteq V_1$, and $q \notin V_1$ by (H2). The valuation V_1 falsifies B_j and hence satisfies A ; it also satisfies Γ by (H1). Since $F \in \mathbf{CL}$, we obtain $r \in V_1$.

Let $H_3 = \{0, \frac{1}{2}, 1\}$ be the three-element Gödel algebra, in which $x \Rightarrow y = 1$ if $x \leq y$ and $x \Rightarrow y = y$ otherwise. Define a valuation v by $v(t) = 1$ if $t \in V_0$, $v(t) = \frac{1}{2}$ if $t \in V_1 \setminus V_0$, and $v(t) = 0$ otherwise. A Horn clause $p_1 \rightarrow \dots \rightarrow p_k \rightarrow q$ takes the value 1 under v if and only if $\min_i v(p_i) \leq v(q)$, where the minimum of the empty set is 1. Now:

- every clause in Γ takes the value 1, since V_0 and V_1 are closed under the rules in Γ ;
- B_j takes the value 0, since $v(p_i) \geq \frac{1}{2}$ for all i and $v(q) = 0$; hence A takes the value 1;
- $v(r) = \frac{1}{2}$, since $r \in V_1 \setminus V_0$.

Therefore F takes the value $\frac{1}{2}$, and $H_3 \not\models F$. By Jankov's criterion [10, Theorem 1] (see also [13, Sect. 3.2]), every implicational formula $F \in \mathbf{CL} \setminus \mathbf{IL}$ with $H_3 \not\models F$ satisfies $\mathbf{IL} + F = \mathbf{CL}$. $\square$

[Plot: Check that [10] or [13, Sect. 3.2] covers the implicational fragment.]

Remark 6.3. In $S(\alpha)$, a positive defining premise $b_u \rightarrow x_u$ has order 3 and a negative one $x_u \rightarrow b_u$ has order 2. Hence the top-level premises of order 3 in $S(\alpha)$ are exactly the defining premises of the positive implication occurrences of α . By Proposition 6.2, if $\alpha \in \mathbf{CL}$ has at most one positive implication occurrence, then $\mathbf{IL} + \alpha = \mathbf{IL} + S(\alpha)$ is either $\mathbf{IL}$ or $\mathbf{CL}$. The formula $S(G')$ has 11 positive defining premises, namely

$$E_1, E_3, E_5, E_7, E_9, E_{11}, E_{12}, E_{14}, E_{15}, E_{17}, E_{20},$$

whereas by Proposition 6.2 a $\mathbf{CL}$ -minimal axiom of order 4 of a proper intermediate logic must have at least two top-level premises of order 3. We do not know the least possible number.

7 Necessity of the Axiom I

In this section we show that the axiom I cannot be dropped from Theorem 1.1; that B and C cannot be dropped from Corollary 5.2 is shown in Appendix C. The results are not used elsewhere in the paper. We write $\mathbf{BC} = \{B, C\}^*$.

The proof of Theorem 1.1 uses I only to discharge the premises $\gamma \rightarrow \gamma$ produced by the decoding (Lemma 4.2). The next proposition shows that this use is essential, even over $\mathbf{BC}$.

Proposition 7.1. *Let $M = (\{0, 1, 2\}, \rightarrow, \{0\})$ be the matrix with*

$\rightarrow$	0	1	2
0	0	1	1
1	0	0	0
2	0	1	1

and let L_M be the set of formulas that take the value 0 under every valuation in M . Then L_M is a logic containing $\mathbf{BC}$ but not I . Moreover, for $\alpha = (p \rightarrow q) \rightarrow (p \rightarrow q)$ the following hold:

- (a) α is minimal in L_M ;
- (b) $S(\alpha) = (x_0 \rightarrow (p \rightarrow q)) \rightarrow ((p \rightarrow q) \rightarrow x_1) \rightarrow ((x_0 \rightarrow x_1) \rightarrow x_\varepsilon) \rightarrow x_\varepsilon$ belongs to L_M ;
- (c) $S(\alpha)$ is not minimal in L_M , since its one-step generalization

$$\alpha' = (x_0 \rightarrow z) \rightarrow (z \rightarrow x_1) \rightarrow ((x_0 \rightarrow x_1) \rightarrow x_\varepsilon) \rightarrow x_\varepsilon,$$

obtained by replacing both bodies $p \rightarrow q$ by z , belongs to L_M .

Hence Theorem 1.1 fails for logics not containing I , even if $S(\alpha) \in L$.

Proof. The set L_M is closed under substitution, and it is closed under modus ponens because $0 \rightarrow y = 0$ implies $y = 0$. A finite check shows that B and C take only the value 0, whereas $2 \rightarrow 2 = 1$; hence $I \notin L_M$.

(a) We have $\alpha \in L_M$, since the values of implications lie in $\{0, 1\}$ and $0 \rightarrow 0 = 1 \rightarrow 1 = 0$. The one-step generalizations of α are

$$\begin{aligned} (z \rightarrow q) \rightarrow (p \rightarrow q), \quad (p \rightarrow z) \rightarrow (p \rightarrow q), \quad (p \rightarrow q) \rightarrow (z \rightarrow q), \quad (p \rightarrow q) \rightarrow (p \rightarrow z), \\ z \rightarrow (p \rightarrow q), \quad (p \rightarrow q) \rightarrow z, \quad z \rightarrow z, \quad z. \end{aligned}$$

Each of them takes a value different from 0 under some valuation in M : for $z \rightarrow z$ take $z = 2$, for z take $z = 1$, and for each of the remaining six formulas a refuting valuation is found by a finite check. Hence α is minimal in L_M by Proposition 2.2.

(b) and (c) are verified by finite checks as well. In fact, α' is the type of the linear λ -term $\lambda f g h. h(\lambda y. g(fy))$, and $S(\alpha)$ is its instance under $z \mapsto p \rightarrow q$; both are theorems of $\mathbf{BC}$. $\square$

Decoding α' as in Lemma 4.2 gives $\beta = z \rightarrow z$, and the premise that would have to be discharged is exactly $z \rightarrow z$, which is not available in L_M .

[Plot: Checks: [tools/bc_alpha_matrix.py](#) (matrix M , found with `z3` and verified by exhaustive evaluation) and [tools/bc_check.py](#) (condensed detachment: $S(\alpha), \alpha' \in \mathbf{BC}$).]

8 Substructural Logics

In this section we extend the translation to richer languages and show that the analogue of Corollary 5.2 holds over $\mathbf{FL}_e$ in the multiplicative language and over $\mathbf{FL}_{ew}$ in the full language, but not over $\mathbf{FL}_e$ with additive connectives. The results of this section are not used elsewhere in the paper.

8.1 The translation in a richer language

Let $\mathcal{L}$ be a language that contains $\rightarrow$ and possibly some of the binary connectives $\cdot$ (fusion), $\wedge$ and $\vee$ and some of the constants $\perp, \top, 0, 1$. An *atom* is a variable or a constant. We consider the full Lambek calculus with exchange $\mathbf{FL}_e$ and its extension $\mathbf{FL}_{ew}$ by weakening, in the language $\{\rightarrow, \cdot, \wedge, \vee, 0, 1\}$, possibly expanded by $\perp$ and $\top$ (see [2]). Intuitionistic logic in this language contains $\mathbf{FL}_{ew}$. The full Lambek calculus $\mathbf{FL}$ without exchange has two implications $\backslash$ and $/$ in place of $\rightarrow$. Logics in $\mathcal{L}$, *one-step generalizations* and occurrences are defined as in Sections 2 and 4, except that an occurrence u is a *leaf* if α_u is an *atom*, and if $\alpha_u = \gamma \circ \delta$ for a binary connective $\circ$, then γ is at $u0$ and δ is at $u1$.

The translation S is defined as in Definition 3.1, with the following changes. If $\alpha_u = \alpha_{u0} \circ \alpha_{u1}$ with $\circ \in \{\cdot, \wedge, \vee\}$, then both $u0$ and $u1$ have the same *polarity* as u . For a *leaf* u we put $r_u = \alpha_u$, so that constants are treated like variables, and the *body* of a compound occurrence u with $\alpha_u = \alpha_{u0} \circ \alpha_{u1}$ is $b_u = r_{u0} \circ r_{u1}$.

Theorem 8.1. *Let L be a logic in $\mathcal{L}$ containing I . If α is minimal in L and $S(\alpha) \in L$, then $S(\alpha)$ is minimal in L .*

Proof. The proof of Theorem 1.1 in Section 4 goes through after the following changes:

- variables of α are replaced by *atoms* of α , except in the condition on δ in the definition of a *one-step generalization*, which still concerns variables only;
- in the “only if” part of the proof of Proposition 2.2 (Appendix A), occurrences of $\rightarrow$ are replaced by occurrences of connectives and constants;
- in Lemmas 4.1(d) and 4.2, the implication in a *body* is replaced by the main connective of the corresponding occurrence. $\square$

8.2 Soundness over $\mathbf{FL}_e$ and $\mathbf{FL}_{ew}$

Lemma 8.2. *In each of the following cases, $\alpha \rightarrow S(\alpha) \in L$ for every formula α of $\mathcal{L}$:*

- (i) $\mathcal{L} \subseteq \{\rightarrow, \cdot, 0, 1\}$ and $L \supseteq \mathbf{FL}_e$;
- (ii) $L \supseteq \mathbf{FL}_{ew}$; in particular, $L \supseteq \mathbf{IL}$.

Proof. We argue as in the proof of Lemma 5.1, in the sequent calculus for $\mathbf{FL}_e$, with weakening in case (ii). The case of fusion is similar to that of implication, and again every formula of Γ_u is used exactly once; this proves (i). For $\wedge$ and $\vee$, the rules $(\wedge R)$ and $(\vee L)$ share their context between the two premises, and in each premise we remove the defining premises belonging to the other subtree by weakening; this proves (ii). $\square$

Corollary 8.3. *In each case of Lemma 8.2, $L + \alpha = L + S(\alpha)$ for every formula α of $\mathcal{L}$. If, moreover, α is minimal in L , then so is $S(\alpha)$.*

Proof. As in the proof of Corollary 5.2, using Lemma 8.2 and Theorem 8.1. $\square$

8.3 Additive connectives over $\mathbf{FL}_e$

The proof of Theorem 8.1 uses only (C1)–(C3), and **polarity** enters only through Lemma 8.2. It is exactly this lemma that fails over $\mathbf{FL}_e$ in the presence of additive connectives. The following two observations, both verified with a cut-free prover for $\mathbf{FL}_e$ (`tools/fle_check.py`), show that the difficulty cannot be removed simply by adding Jeřábek’s wrappers “ $\wedge 1$ ”.

Remark 8.4. Let $\mathcal{L} = \{\rightarrow, \cdot, \wedge, \vee, 0, 1\}$.

- (a) *Without the wrappers, S is not sound over $\mathbf{FL}_e$.* The formula $\alpha = (q \rightarrow q) \wedge (r \rightarrow r)$ is minimal in $\mathbf{FL}_e$, but

$$S(\alpha) = ((q \rightarrow q) \rightarrow x_0) \rightarrow ((r \rightarrow r) \rightarrow x_1) \rightarrow ((x_0 \wedge x_1) \rightarrow x_\varepsilon) \rightarrow x_\varepsilon$$

is not provable in $\mathbf{FL}_e$: each premise of the rule $(\wedge R)$ would have to discard the defining premise belonging to the other conjunct.

- (b) *With the wrappers, minimality is always lost.* Jeřábek’s φ^+ has $E_u \wedge 1$ in place of E_u , which makes it sound over $\mathbf{FL}_e$ [3, Thm. 3.8]. The defining premise of the root is never discarded in the proof, so replacing its conjunct 1 by z gives a one-step generalization that is still provable, by $(\wedge L)$. Hence φ^+ is not minimal in any logic $L \supseteq \mathbf{FL}_e$, including $\mathbf{FL}_{ew}$, $\mathbf{IL}$ and $\mathbf{CL}$, whenever φ is compound. For example, $((p \rightarrow p) \rightarrow x_\varepsilon) \wedge z \rightarrow x_\varepsilon$ is provable.

Exhaustive tests over all formulas in $\rightarrow, \wedge, \vee$ with at most four binary connectives and three variables give the following picture. Up to renaming, there are 413 formulas minimal in $\mathbf{FL}_e$. For 164 of them $S(\alpha)$ is provable, and in all these cases it is minimal, as predicted by Theorem 8.1. The formula φ^+ is provable in all 413 cases and minimal in none. Wrapping only the defining premises of occurrences strictly below an additive occurrence yields a provable formula in all 413 cases, but a minimal one in only 255. For $\alpha = p \vee (q \rightarrow q)$, for instance, the wrapper can be removed because the rule $(\vee R)$ selects one disjunct. Thus a minimality-preserving translation over $\mathbf{FL}_e$, if it exists, must adapt the wrappers to the actual proof. We leave open whether such a translation exists, as well as the non-commutative case $\mathbf{FL}$.

9 Concluding Remarks

We have shown that a variant of Jeřábek’s polarized extension-variable translation, without the wrappers “ $\wedge 1$ ”, preserves minimality with respect to the substitution preorder in every logic that contains the axiom I and the translated formula. As a consequence, axiomatizations of implicational logics over $\mathbf{BCI}$ by $\mathbf{CL}$ -minimal formulas can always be taken to consist of formulas of order at most 4, and [13, Open Problem 2] has an affirmative answer. We conclude with some questions that remain open.

1. Which intermediate logics are axiomatizable by **CL**-minimal formulas [13, Open Problem 1]? By Corollary 5.3 and Proposition 5.4, it suffices to consider formulas of order 4.
2. Is there a minimality-preserving translation over **FL_e** with additive connectives, or over the non-commutative calculus **FL** (Section 8.3)?
3. Does the translation, possibly with reordered premises, preserve minimality over ticket entailment **T**_→ (Appendix C)?
4. What is the least number of top-level premises of order 3 in a **CL**-minimal axiom of order 4 of a proper intermediate logic? By Proposition 6.2 it is at least 2, whereas **S**(*G'*) has 11 such premises.

Acknowledgements. [\[Plot: JSPS KAKENHI etc.\]](#)

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A Proof of Proposition 2.2

Proof of Proposition 2.2. “Only if”. Suppose that α is minimal in L . Let $\beta = \alpha[Q:=z]$ be a one-step generalization of α ; we show that $\beta <_{\text{sub}} \alpha$, so that $\beta \notin L$. Since z does not occur in α , substituting δ for z in β yields α ; hence $\beta \leq_{\text{sub}} \alpha$. Suppose, towards a contradiction, that $\tau\alpha = \beta$ for some substitution τ . Note that a substitution never decreases the number of occurrences of $\rightarrow$.

Case: δ is not a variable. Then δ contains $\rightarrow$, so β has fewer occurrences of $\rightarrow$ than α , which is impossible.

Case: δ is a variable p . Then α and β have the same number of occurrences of $\rightarrow$, so τ maps each variable of α to a variable. Thus $\tau\alpha$ has the same syntax tree as α and carries the variable $\tau(p)$ at every leaf at which α carries p . At these leaves, however, β carries z (at the elements of Q) as well as p (at an occurrence outside Q). Hence $\tau(p) = z$ and $\tau(p) = p$, a contradiction.

“If”. Suppose that no one-step generalization of α belongs to L . Let $\gamma \in L$ and let σ be a substitution with $\sigma\gamma = \alpha$. We show that $\alpha \leq_{\text{sub}} \gamma$.

Case: σ maps the variables of γ injectively to variables. Then there is a substitution τ with $\tau(\sigma(q)) = q$ for every variable q of γ , and $\tau\alpha = \tau\sigma\gamma = \gamma$.

Case: otherwise. There is a variable q of γ such that either (i) $\sigma(q)$ is not a variable, or (ii) $\sigma(q) = \sigma(q')$ is a variable for some variable $q' \neq q$ of γ . Let z be a variable not occurring in α , and let σ' agree with σ except that $\sigma'(q) = z$. Every occurrence of q in γ gives rise to an occurrence of $\delta := \sigma(q)$ in $\alpha = \sigma\gamma$; let Q be the set of these occurrences. Then $Q \neq \emptyset$ and $\sigma'\gamma = \alpha[Q:=z]$. In case (ii), the occurrences of q' in γ give rise to occurrences of the variable δ that do not belong to Q . Therefore $\sigma'\gamma$ is a one-step generalization of α , and $\sigma'\gamma \in L$ by (C1), contrary to the hypothesis. $\square$

B Formalization in Lean

[Plot: Half a page.]

Theorem 1.1, Lemma 5.1, the `order` bound of Corollary 5.3, Corollary 5.2 and the explicit formula of Corollary 6.1 have been formalized in the Lean 4 proof assistant [15]. The development consists of a single file, `lean/StatmanMinimality.lean`, of about 1300 lines. It uses only the Lean core library, without Mathlib, and contains neither `sorry` nor `native_decide`; the command `#print axioms` reports only the standard axioms `propext`, `Classical.choice` and `Quot.sound`. (The identifiers in the formalization use the name `statman` for the translation S .)

Theorem 1.1 is formalized in exactly the generality stated. A logic is represented by a predicate `Th` on formulas satisfying `PreLogic Th`, which expresses closure under substitution and modus ponens together with all instances $\gamma \rightarrow \gamma$ of I . The main theorem reads

```
statman_minimalIn : PreLogic Th →
  (∀ q, Occ q A → q < N) → MinimalIn Th A →
  Th (statman A N) → MinimalIn Th (statman A N),
```

where the hypothesis on N ensures that the extension variables are fresh. The classical instance `statman_minimal` is derived from it by means of `logic_valid` and `statman_valid`.

Lemma 5.1 and the decoding argument in the proof of Corollary 5.2 are formalized for every logic containing **BCI**, represented by `BCIExt Th` (closure under substitution and modus ponens, and all instances of B , C and I), as

```
BCIExt.statman_imp : Th (A → statman A N),
BCIExt.statman_iff : (∀ q, Occ q A → q < N) →
  (Th A ↔ Th (statman A N)),
```

from which the equation $L + \alpha = L + S(\alpha)$ of Corollary 5.2 follows.

The theorem `BCIExt.statman_minimal` is Corollary 5.2, without further hypotheses. We have also formalized the `order` bound $\text{Ord}(S(\alpha)) \leq 4$ as `statman_ord`, and the explicit formula of Corollary 6.1: `statman_Gprime` states that the displayed formula $S(G')$ equals `statman_Gprime` 8, and `SGprime_ord` states that it has order 4. The results of Section 8 have not been formalized.

The formal proof follows the paper closely. The lemmas `reduction` and `strict` correspond to Proposition 2.2, and `decode` and `decode_node` correspond to Lemma 4.2; the latter are organized node by node rather than via the reduction to basic generalizations of Lemma 4.3.

C Necessity of the Axioms B and C

Theorem 1.1 requires that $S(\alpha) \in L$, which is guaranteed over **BCI** by Lemma 5.1. We show $\sqcap$ that the axioms B and C cannot be dropped here. We use the logics

$$\mathbf{CK} = \{C, K\}^*, \quad \mathbf{BK} = \{B, K\}^*, \quad \mathbf{BB'KI} = \{B, B', K, I\}^*$$

and ticket entailment $\mathbf{T}_{\rightarrow} = \{B, B', I, W\}^*$.

Proposition C.1. *The following hold.*

(a) *(Without B .) The formula K is minimal in $\mathbf{CK}$, but*

$$S(K) = ((b \rightarrow a) \rightarrow x_1) \rightarrow ((a \rightarrow x_1) \rightarrow x_\varepsilon) \rightarrow x_\varepsilon$$

does not belong to $\mathbf{CK}$.

- (b) (Without C .) The formula I is minimal in $\mathbf{BB'KI}$, but $S(I) = ((a \rightarrow a) \rightarrow x_\varepsilon) \rightarrow x_\varepsilon$ does not belong to $\mathbf{BB'KI}$. Likewise, K is minimal in $\mathbf{BK}$ and in $\mathbf{BB'KI}$, but $S(K) \notin \mathbf{BB'KI} \supseteq \mathbf{BK}$.

Consequently, if L is $\mathbf{CK}$ or a logic between $\mathbf{BK}$ and $\mathbf{BB'KI}$, then the translation S does not preserve minimality in L , and $L + \alpha \neq L + S(\alpha)$ for $\alpha = K$; for $L = \mathbf{BB'KI}$ the same holds also for $\alpha = I$.

Proof. Minimality. The one-step generalizations of K are $z \rightarrow b \rightarrow a$, $a \rightarrow b \rightarrow z$, $a \rightarrow z$ and z , and those of I are $z \rightarrow a$, $a \rightarrow z$ and z . None of them is classically valid, so none of them belongs to any of the logics under consideration, all of which are contained in $\mathbf{CL}$.

- (a) Let $M_4 = (\{0, 1, 2, 3\}, \rightarrow, \{0\})$ be the matrix with

$\rightarrow$	0	1	2	3
0	0	1	2	3
1	0	0	1	0
2	0	0	0	3
3	0	1	2	0

A finite check shows that M_4 validates C and K (and hence also I , which is derivable in $\mathbf{CK}$) but not B . The set $\{0\}$ is closed under modus ponens, since $0 \rightarrow y = 0$ implies $y = 0$, so every theorem of $\mathbf{CK}$ takes the value 0. However, under the valuation $a = b = 1$, $x_1 = 2$ and $x_\varepsilon = 3$ we have $b \rightarrow a = 0$, $(b \rightarrow a) \rightarrow x_1 = 2$, $a \rightarrow x_1 = 1$ and $(a \rightarrow x_1) \rightarrow x_\varepsilon = 0$, so $S(K)$ takes the value $2 \rightarrow (0 \rightarrow 3) = 2 \rightarrow 3 = 3 \neq 0$.

(b) Let $M_3 = (\{0, 1, 2\}, \rightarrow, \{0\})$, where $x \rightarrow y = 0$ if $y \leq x$ and $x \rightarrow y = 1$ otherwise. A finite check shows that M_3 validates B , B' , K and I . As before, $\{0\}$ is closed under modus ponens, so every theorem of $\mathbf{BB'KI}$ takes the value 0 in M_3 . However, under the valuation $a = 0$ and $x_\varepsilon = 2$, the formula $S(I)$ takes the value $(0 \rightarrow 2) \rightarrow 2 = 1 \rightarrow 2 = 1$. The formula $S(K)$ is refuted similarly. (The matrix M_3 refutes C and W .) $\square$

Ticket entailment $\mathbf{T}_\rightarrow$ contains neither C nor K , and the matrix M_3 does not apply to it, since M_3 refutes W . We do not know whether $S(\alpha) \in \mathbf{T}_\rightarrow$ for every $\alpha \in \mathbf{T}_\rightarrow$.

[Plot: Checks: `tools/ck_z3.py` (matrix M_4 , found with `z3` and verified by exhaustive evaluation), `tools/bki_matrix.py` (matrix M_3), `tools/ck_cd.py` and `tools/bk_cd.py` (condensed-detachment search, consistent with (a) and (b)). Open: ticket entailment $\mathbf{T}_\rightarrow$, where M_3 does not apply because it refutes W ; and whether reordering the premises helps. For (b), every order of premises fails already for $S(I)$, which has a single premise.]